Ziyang Liu

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Education

National University of Singapore Aug 2026 – Nov 2027
M.Tech. in Software Engineering *Singapore*

University of Electronic Science and Technology of China Sep 2020 – Jun 2024
B.Eng. in Software Engineering *Chengdu, China*

Experience

Virtualization R&D Engineer Jul 2024 – Sep 2025
SmartX *Chengdu, China*

- **Migration product:** led the design of V2V OS 1.6.0 to 1.7.2, which moves virtual machines from VMware to SMTX OS and between SMTX clusters. Built across the end-to-end path: VMware CBT snapshots, a Python control plane, and a C data-transfer core exposed as a shared library. The product now serves 10,000+ production VMs, including core financial workloads.
- **Transfer performance:** replaced the serial read-then-write path with asynchronous chunked reads and writes across a bounded queue, using 8 reader and 8 writer workers. Handled partial I/O, cancellation and first-error propagation across the pipeline. Increased sustained throughput from 70 MB/s to 290 MB/s.
- **VMware compatibility:** isolated each migration in its own process, with one VDDK version loaded per process. This removed shared-process SDK symbol and global-state conflicts, contained crashes to one task, and expanded support across vSphere releases.
- **Destination-platform support:** prepared migrated VMs to boot on new virtual hardware through offline driver injection. Replaced the NFS disk-attachment path after hung mounts blocked guest preparation, using iSCSI block devices instead; then adapted the pipeline to SMTX ELF disaggregated compute-storage clusters.
- **In-guest agent:** rebuilt VMTools, replacing shell scripts over QGA guest-exec with native QEMU Guest Agent commands and resource probes. Cut response time from 3 s to under 50 ms and yearly escalations from 40 to under 5.
- **Consistent snapshots:** made snapshots filesystem-consistent rather than only crash-consistent. The control plane coordinates VMTools and fsfreeze to quiesce the guest filesystem before the storage snapshot, then thaws it after the snapshot boundary.
- **Cross-cluster live migration:** built a block-level data check triggered by a libvirt hook after switchover. It snapshots source and destination disks through the storage API, reads and hashes them block by block, and keeps compare order in sequence queues on both sides so verification runs concurrently without blocking migration.
- **Kernel and QGA debugging:** used strace to find a UOS kernel creating pidfds without reaping them, hanging customer VMs; the fix became a financial-sector advisory and earned a written commendation. Also traced truncated host-to-guest JSON to old kernels not re-polling the QGA serial fd.

Virtualization R&D Intern Oct 2022 – Jul 2024
SmartX *Chengdu, China*

- **Migration appliance:** built V2V OS 1.2.0 to 1.6.0 as a self-contained virtual appliance on openEuler for government procurement. Rebuilt and packaged its kernel with the filesystem and block-device features required to inspect and modify newer guest images.
- **Low-downtime migration:** extended the workflow from one full copy plus a long outage to an initial copy followed by repeated incremental passes while the VM keeps running. VMware Changed Block Tracking (CBT) identifies blocks dirtied after each snapshot, leaving only a small final delta for the stopped-VM switchover.
- **Packaging and release:** wrote the rpm.spec for V2V and its custom kernel. Built CI/CD that assembles both into a bootable ISO and a Packer pipeline that produces the vSphere OVF appliance.
- **Offline driver injection:** made migrated guests boot on virtio hardware by rebuilding Linux initramfs images and replaying Windows registry changes captured with procmon around a virtio-win installation. Supported both MBR and UEFI, with disk snapshots and an IDE fallback for recovery.
- **Guest coverage:** carried driver injection and VMTools across nine Linux distributions, including openEuler, UOS and Kylin, plus Windows Server 2012–2025. Maintained per-OS handling for boot layout, package tools, service startup and guest-agent installation.

Systems Integration Development Intern Jun 2022 – Sep 2022
ByteDance (Feishu) *Chengdu, China*

- **Workflow integration:** built Go and Gin services connecting recruiting workflows to external OA systems, with MongoDB holding sync and retry state.
- **API hardening:** added authentication, rate limiting and a regression suite across the integration endpoints.

Project

Maintainer Sep 2026 – Present

FastMM, market-making engine in C++20 *ziy.bio/FastMM*

- **Live results:** in Binance's bStock Spot Liquidity Provider Program, made markets on 76 tokenized US-stock pairs with about $36K. Traded **$23.28M in 30 days**, with a peak week of $7.53M. Reached the **top rebate tier, Tier 4** (maker fee −0.008%). **#1 on 2 pairs** in each of two consecutive weeks, up to 35% share of one pair, and top 3 on 13 pairs.
- **Hot path:** designed the engine around one trading thread that owns all state, so the hot path needs no locks; other threads reach it through lock-free queues. No memory is allocated after start-up. In the simulator benchmark, market data to order takes 159 ns.
- **Live latency fix:** found orders waiting 2 to 6 ms inside the engine in live trading, because each balance update re-priced all 85 pairs. Changed re-pricing to once per batch, cutting balance handling p99 from 23.8 to 1.2 µs.
- **CPU contention:** found six busy-waiting threads competing for the 2-vCPU live-trading cloud host. Switched idle threads from spinning to backing off, cutting latency p50 from 2.5 to 0.23 ms and p99 from 7.0 to 0.49 ms.
- **Market data:** switched Binance market data from JSON to its binary format, cutting decode time from 680 to 51 ns. Wrote connectors for 7 crypto exchanges and Nasdaq, plus AF_XDP and DPDK receive paths.
- **Backtest accuracy:** made backtest and live run the same strategy code, with deterministic replays. Replayed with real arrival times and latency, raising backtest adverse fills from 2.6% to 16.0%, against 17.2% live.
- **Python:** added a Python strategy interface, with Numba compiling the fast parts to machine code. Published it on PyPI as fastmm-engine.
- **Dashboard:** built bndesk (source, demo), a Rust dashboard for live trading: P&L by source, fill quality, quotes and inventory for every sub-account. Made the server send only what changed. Live demo on request.

Research

- **Disproved a linear-algebra conjecture:** built a 3×3 counterexample to a 2017 matrix-approximation conjecture; proved that it holds up to 2×2 and that the approximation always converges.
  Limits of Schatten Best Approximants: Universal Convergence and Failure of Strict Spectral Selection.
  ***Z. Liu**, Y. Wang, Y. Shi*
- **Solved two open problems in number theory:** assuming the Generalized Riemann Hypothesis, solved two problems from a 2024 list in comparative prime number theory; for one, computed the exact share of all 120 orderings of five prime classes.
  Two Problems in Comparative Prime Number Theory: The Five-Way Race Modulo 11, and the Sign of the Integrated Prime Counting Error. ***Z. Liu**, Y. Wang, Y. Shi*
- **In-context learning curves:** showed that the standard curve misleads: average loss falls but rises within every token-frequency group.
  A Simpson's Paradox in In-Context Learning Curves. ***Z. Liu**, Y. Shi. ICLR 2027 submission*
- **Model identity audit:** detects a hosted LLM API that swaps in a cheaper model; caught all 17 substitutes.
  Committed SAE-Feature Traces for Audited-Session Substitution Detection in Hosted LLMs. ***Z. Liu***
- **Long-conversation memory:** lets the model recall text that has left the context window; beat 5 methods on LoCoMo.
  Cooperative Memory Paging with Keyword Bookmarks for Long-Horizon LLM Conversations. ***Z. Liu***
- **Faster decoding for edits:** lets an LLM copy unchanged text while editing a file; speedup bound of 13×.
  Copy-as-Decode: Grammar-Constrained Parallel Prefill for LLM Editing. ***Z. Liu***
- **AI-generated video benchmark:** on 21,504 real platform videos, published detectors fall from 93–96% to 52–71% AUROC, near chance.
  A Living In-the-Wild Benchmark for Measuring the Static-Benchmark Gap in AI-Generated Video Detection.
  Y. Shi, ***Z. Liu**, K. Luo, N. Tang. ICLR 2027 submission*
- **Training-free video detection:** reaches 97.39% accuracy on 10 unseen video generators.
  Neuro-Symbolic Forensic Reasoning for Open-Generator AI-Generated Video Detection.
  Y. Shi, K. Luo, ***Z. Liu**, N. Tang. ICLR 2027 submission*

Technical Skills

Languages: Python, C++, C, Rust, Go, Shell
Networking: TCP/UDP multicast, epoll, io_uring, AF_XDP, DPDK, tcpdump
Linux/UNIX: processes and threads, virtual memory, NUMA, CPU affinity, cgroups, strace/perf/gdb
Virtualisation & migration: QEMU/KVM, libvirt, Virtio, VMware CBT/VDDK, QGA, live migration
LLM systems: KV cache, speculative decoding, SAE interpretability, long-context memory, reranking, ReAct/MCP
Build & data: CMake, GitHub Actions, Jenkins, Docker, RPM/DEB, Packer, MySQL, Redis, MongoDB

Honours

National Third Prize, China Collegiate Computer Design Competition (2022) | UESTC Scholarship (2020–21)